as puravuvari-tinaikkalam, with its chief called 'puravuvari-tinaikkalanayagam'.

Land Revenue and Survey

For the purposes of assessing tax, the Cholas undertook extensive land surveys and revenue settlements. Rajaraja I (1001), Kulotunga I (1086) and Kulotunga III (1226) appointed people for land survey so that the land could be classified and assessed for the purposes of taxation. Like other functionaries of the state, the surveyors of the land called naduvagaiseykira too hailed from the landholding communities. Various units of the land measurement such as kuli, ma, veli, patti, padagam, etc. are known, with local variations. Generally, taxes were collected in different forms. The taxes collected included irai, kanikadan, iraikattina-kanikadan and kadamai. An important category of tax was kudimai. Kudimai was paid by the cultivating tenants to the government and to the landlords, the bearers of honorific titles such as udaiyan, araiyan and kilavar. The tax rates were fixed depending on the fertility of the soil and the status of the landholder. Opati were levied and collected by the king and local chiefs. Temples and Brahmins were exempted from paying the taxes. The tax paid in kind was referred to as iraikattina-nellu. All these were mostly realised from the Kavery delta but not widely in the outskirts of the kingdom. At the ur (village) level, urar (village assembly) were responsible for collecting the taxes and remitting them to the government. At the nadu level, the nattar were responsible for remitting taxes.

Paddy as tax was collected by a unit called kalam (28 kg). Rajaraja I standardised the collection of tax. He collected 100 kalam from the land of one veli (about 6.5 acres), the standard veli being variable according to fertility of the soil and the number of crops raised.

Irrigation

Cholas undertook measures to improve the irrigation system that was in practice. As the state was drawing most of its revenue from agriculture, the Cholas focused their efforts on managing water resources. Vativaykkal, a criss-cross channel, is a traditional way of harnessing rain water in the Kavery delta. Vati runs in the north–south direction while vaykkal runs in the east–west direction. Technically, vati is a drainage channel and a vaykkal is a supply channel. The water running through vaykkal to the field was to be drained out to vati and to another vaykkal. Rain water would flow from where the natural canal started. Many irrigation canals are modifications of such natural canals. The harnessed water was utilised alternately through vati and vaykkal. Here the mechanism designed was such that water was distributed to the parcelled out lands in sequel.

Many canals were named after the kings, queens and gods. Some examples of the names are Uttamachola-vaykkal, Panca-vanamadevi-vaykkal and Ganavathy-vaykkal. Ur-vaykkal was owned jointly by the landowners. The nadu level vaykkal was referred to as nattu-vaykkal. The turn system was practiced for distributing the water. Chola inscriptions list some big-size irrigation tanks such as Cholavaridhi, Kaliyaneri, Vairamegatataka created by the Pallavas, Bahur big tank and Rajendra Cholaperiyaeri. For the periodical or seasonal maintenance and repair of irrigation works, conscripted labour was used.

The irrigation work done by Rajendra Chola I at Gangaikonda Chozhapuram was an embankment of solid masonry 16 miles long. Rajendra described it as his 'jalamaya jayasthambham', meaning "pillar of victory in water". The Arab traveller Alberuni visited the place a hundred years later. On seeing them he was wonder-struck and said: "'Our people, when they see them, wonder at them, and are unable to describe them, much less construct anything like them", records Jawaharlal Nehru in *The Glimpses of World History.*

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